Oracle Data Guard Physical Standby

Most Common Things Every DBA Must Keep in Mind

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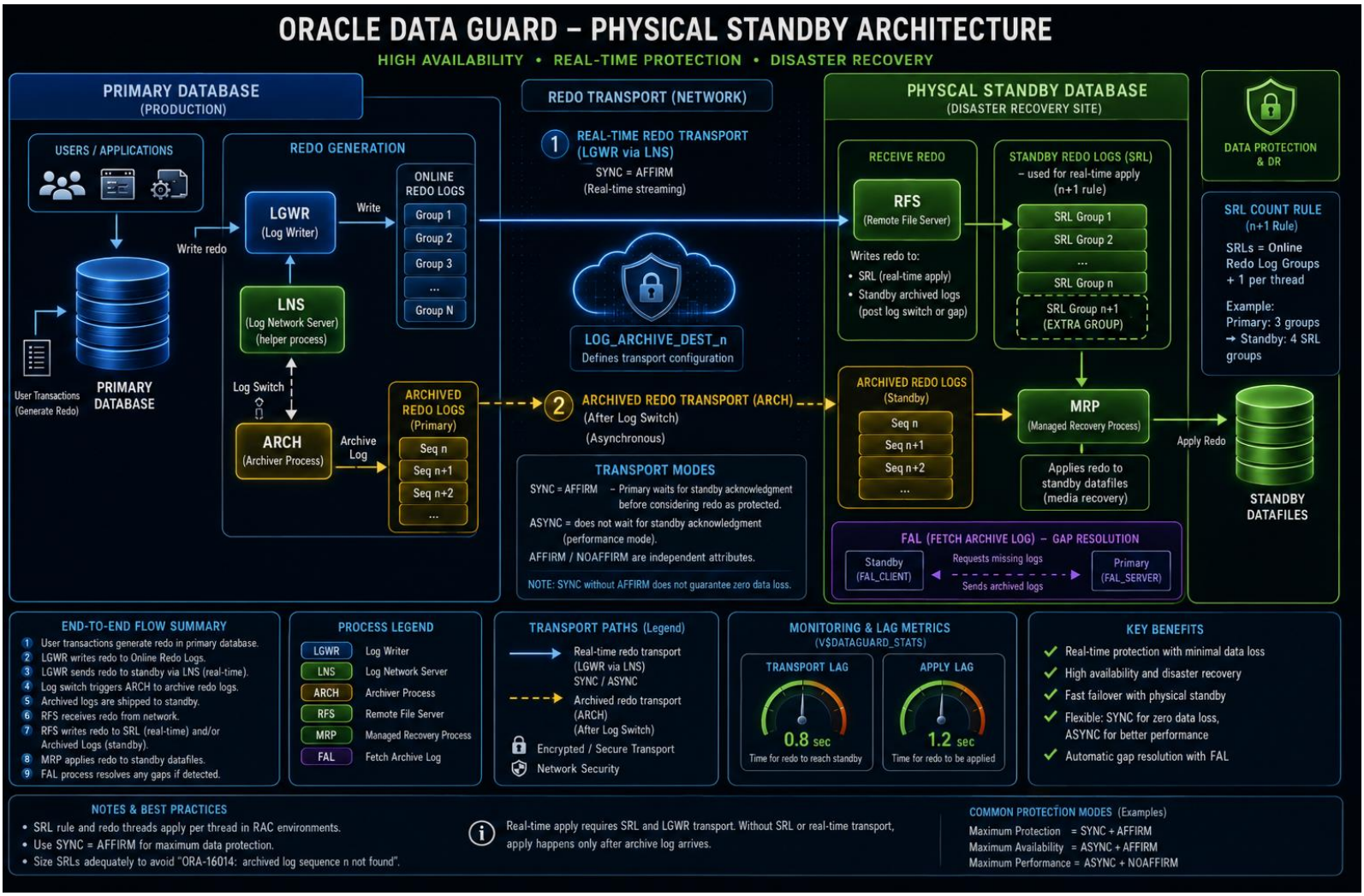


Table of Contents

1. Introduction to Oracle Data Guard
2. What is Physical Standby Database
3. Primary vs Physical Standby Architecture
4. Redo Transport Services
5. Redo Apply Process
6. Data Guard Components and Background Processes
7. ARCH vs LGWR Transport Modes
8. SYNC vs ASYNC Transport
9. Protection Modes: Maximum Protection, Availability, and Performance
10. Standby Redo Logs Best Practices
11. Real-Time Apply
12. Log Apply Services
13. Data Guard Broker Overview
14. Important Initialization Parameters
15. Network Configuration Best Practices
16. TNS and Listener Configuration
17. Password File Management
18. RMAN Duplication for Standby
19. Active Duplicate Method
20. Monitoring Physical Standby Health
21. Important Monitoring Queries
22. Gap Detection and Resolution
23. Archive Log Issues
24. FAL_SERVER and FAL_CLIENT
25. Apply Lag and Transport Lag
26. Switchover Process
27. Failover Process
28. Fast-Start Failover Concepts
29. Data Guard Broker Commands
30. Common Production Issues
31. ORA Errors Commonly Seen in Data Guard
32. Troubleshooting Real-Time Examples
33. Recovery Scenarios
34. Standby Database Maintenance
35. Patch and Upgrade Considerations
36. Backup Strategy in Data Guard Environment
37. RMAN Best Practices
38. Performance Tuning in Data Guard
39. Common Mistakes DBAs Make
40. Important Production Checklists
41. Daily Health Check Activities

42. DR Drill Planning
43. Best Practices in Real Production Environments
44. Interview-Focused Practical Concepts
45. Final Summary and Key Takeaways

1. Introduction to Oracle Data Guard

Oracle Data Guard is Oracle's built-in high availability, data protection, and disaster recovery solution. It maintains one or more synchronized copies of a production database at a remote or local location. These copies — called standby databases — are kept in sync with the primary by continuously shipping and applying redo data.

Data Guard is not just a backup — it is an active protection layer that ensures your database survives disk failures, site outages, human errors, and even data corruption. In a production environment, running without Data Guard is considered a serious risk.

The idea is simple: every change made on the primary database generates redo. That redo gets shipped to the standby, which applies it continuously. When disaster hits — whether planned or unplanned — you can switch roles in minutes, or sometimes seconds.

Why Data Guard Matters in Production

- Zero data loss is achievable with synchronous redo transport
- Standby databases can offload read workloads (Active Data Guard license required)
- Switchover and failover are clean, tested operations — not emergency improvisation
- Backup offload to standby reduces load on the primary
- Rolling upgrades are possible using a physical standby

Things to Remember:

Data Guard is not a replacement for RMAN backups. Use both. Data Guard protects against site failures; RMAN protects against logical corruption and user errors.

2. What is Physical Standby Database

A physical standby is a block-for-block identical copy of the primary database. It is maintained through Redo Apply — Oracle applies redo data from the primary directly to the standby datafiles using media recovery. At any point in time, the standby is physically identical to the primary up to the last applied redo.

This is different from a logical standby (which uses SQL Apply and runs open for reads). A physical standby is the most commonly used standby type in production environments because it is simpler, faster, and supports all datatypes and DDL operations without restrictions.

Physical Standby States

State	Description
MOUNTED	Database is mounted but not open. Redo Apply can run in this state.
READ ONLY	Database is open for reads. Active Data Guard license required to apply redo while open.
READ ONLY WITH APPLY	Open for reads AND applying redo simultaneously (Active Data Guard).

Most production environments run the physical standby in MOUNT state (applying redo) unless they have Active Data Guard, in which case READ ONLY WITH APPLY provides real-time readable copies.

Production Best Practice:
Always keep the physical standby in a state where Redo Apply is running. A standby that is not applying is falling behind and provides no guarantee of data freshness at failover time.

3. Primary vs Physical Standby Architecture

Understanding the architecture is the first step to troubleshooting and managing Data Guard confidently. Here is how it works from end to end:

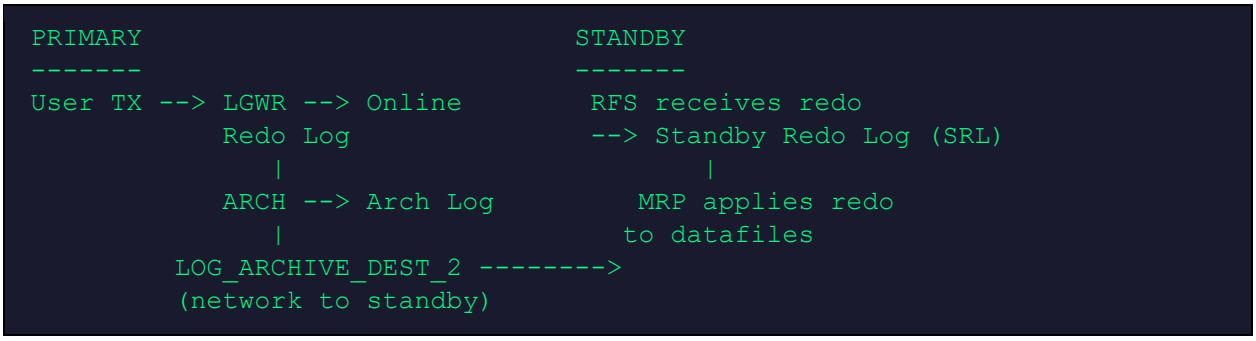
Primary Database Side

- Users commit transactions. Oracle generates redo in the online redo logs.
- LGWR (Log Writer) writes redo to the online redo log group currently active.
- When a redo log switch happens, ARCH (Archiver) archives the filled log to the archive log destination.
- In LGWR transport mode, redo is shipped by LGWR directly to the standby in real-time, even before archiving.
- The standby destination is defined in LOG_ARCHIVE_DEST_n parameters.

Standby Database Side

- The RFS (Remote File Server) process on the standby receives redo from the primary.
- Received redo is written into Standby Redo Logs (SRLs) — these are mandatory for real-time apply.
- MRP (Managed Recovery Process) reads the redo from SRLs (or archived logs if SRLs are not configured) and applies it to the standby datafiles.
- When a log switch occurs on the primary, the SRL on the standby is archived, and MRP moves to the next sequence.

Architecture Flow Summary



Things to Remember:
If Standby Redo Logs are NOT configured, the standby will only apply redo after log switches (after the archive is received). This introduces more lag. Always configure SRLs.

4. Redo Transport Services

Redo transport is the mechanism that moves redo data from the primary database to the standby. Getting this configured correctly is fundamental — every problem in Data Guard eventually traces back to transport or apply.

Key Parameters for Redo Transport

```
-- On Primary Database
LOG_ARCHIVE_DEST_1 = 'LOCATION=/u01/arch
VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=PROD'
LOG_ARCHIVE_DEST_2 = 'SERVICE=STDBY ASYNC
VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
                      DB_UNIQUE_NAME=STDBY LGWR'
LOG_ARCHIVE_DEST_STATE_1 = ENABLE
LOG_ARCHIVE_DEST_STATE_2 = ENABLE
```

Breaking down the key attributes in LOG_ARCHIVE_DEST_n:

Attribute	Purpose
SERVICE	TNS alias of the standby database listener
ASYNC / SYNC	Transport mode — async does not wait for standby acknowledgment; sync waits
LGWR / ARCH	Which process sends redo — LGWR sends in real-time; ARCH sends after log switch
DB_UNIQUE_NAME	Unique name of the target standby database
VALID_FOR	Defines when this destination is active (role and logfile type)
AFFIRM / NOAFFIRM	Whether standby must write to disk before acknowledging

5. Redo Apply Process

Redo Apply is the heart of physical standby operation. The MRP process takes the received redo and applies it to the standby datafiles, making the standby mirror the primary.

Starting and Stopping MRP

```
-- Start Redo Apply in background (most common):
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT FROM SESSION;

-- Start with Real-Time Apply (uses SRLs, applies without waiting for log
switch):
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;

-- Stop Redo Apply:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

-- Check MRP status:
SELECT PROCESS, STATUS, SEQUENCE#, THREAD# FROM V$MANAGED_STANDBY;
```

MRP Process States

Status	Meaning
APPLYING_LOG	MRP is actively applying redo — this is the healthy state
WAIT_FOR_LOG	MRP is waiting for the next log to arrive from primary
WAIT_FOR_GAP	A gap in archive log sequences has been detected
IDLE	MRP is running but not applying — check if redo is being received

Things to Remember:
WAIT_FOR_LOG is normal if the primary has low activity. WAIT_FOR_GAP means you have a gap problem — investigate immediately. Check V$ARCHIVE_GAP on the primary.

6. Data Guard Components and Background Processes

Knowing which process does what makes it much easier to diagnose issues quickly. Here are all the key processes involved in Data Guard:

Primary Side Processes

Process	Role
LGWR	Writes redo to online redo logs. In LGWR transport mode, also ships redo to standby in real-time
ARCH	Archives filled online redo logs. In ARCH transport mode, ships archives to standby
LNS (Log Network Server)	In LGWR mode, LNS assists LGWR in sending redo over the network to the standby
DMON	Data Guard Monitor — part of the broker framework, runs when broker is enabled

Standby Side Processes

Process	Role
RFS (Remote File Server)	Receives redo from primary and writes it to Standby Redo Logs or archived log destinations
MRP (Managed Recovery Process)	Applies redo to standby datafiles. This is what keeps the standby in sync
LSP (Log Sync Process)	Assists MRP with real-time apply when using current SRLs
FAL (Fetch Archive Log)	Fetches missing archive log sequences from the primary to resolve gaps
DMON	Broker monitor on standby side, handles role transitions and health checks

```
-- Check all Data Guard processes on standby:
SELECT PROCESS, STATUS, THREAD#, SEQUENCE#, BLOCK#, BLOCKS
FROM V$MANAGED_STANDBY
ORDER BY PROCESS;
```

7. ARCH vs LGWR Transport Modes

One of the most commonly asked questions — and most misunderstood concepts — in Data Guard is the difference between ARCH and LGWR transport. The choice of transport process determines latency, performance impact, and how much data you can potentially lose.

ARCH Transport

In ARCH mode, the archiver process sends redo to the standby AFTER a redo log switch occurs. This means the standby is always at least one redo log behind. If the primary crashes between log switches, the current unsent redo in the online log is lost.

- Lower impact on primary transactions
- Suitable for environments where some data loss is acceptable
- The ARCH process ships the archive log file once created
- Used with ASYNC parameter in LOG_ARCHIVE_DEST_n

LGWR Transport

In LGWR mode, the Log Writer process ships redo to the standby in real-time as it writes to the online redo log. The standby receives redo almost simultaneously with the commit. This is the preferred mode for production environments requiring minimal data loss.

- LGWR spawns a separate process (LNS - Log Network Server) to handle the network I/O
- Supports both SYNC and ASYNC sub-modes
- Required for SYNC transport (zero data loss)
- Higher network dependency — network issues directly affect primary performance with SYNC

Quick Comparison

Feature	ARCH Mode	LGWR Mode
When redo is sent	After log switch	Real-time (per commit)
Data loss risk	Up to one log file	Minimal / Zero with SYNC
Primary performance impact	Low	Low (ASYNC) / Possible (SYNC)
Standby lag	Higher	Very low
Required for SYNC	No	Yes

Common DBA Mistake:
Never assume ARCH mode provides the same protection as LGWR. In a primary crash, everything since the last log switch is at risk. For production databases with critical transactions, always use LGWR.

8. SYNC vs ASYNC Transport

After deciding on LGWR transport, you need to choose between SYNC and ASYNC. This decision has a direct impact on primary database performance and the level of data protection you achieve.

ASYNC (Asynchronous)

With ASYNC, the primary does not wait for the standby to acknowledge receipt of redo before completing a commit. The LNS process sends redo to the standby in the background, but the primary user transaction is not held.

- No performance impact on primary transactions
- Small window of data loss possible if primary fails and LNS buffer has unsent redo
- Suitable for long-distance standby or when network latency is high
- Used with Maximum Performance and Maximum Availability protection modes

SYNC (Synchronous)

With SYNC, the primary waits for the standby to confirm it has received the redo before the commit completes. This guarantees zero data loss — the standby has received everything the primary has committed.

- Zero data loss — no committed transaction is ever lost
- Primary commit performance is tied to network round-trip time to standby
- If the standby or network becomes unreachable, behavior depends on protection mode setting
- Required for Maximum Protection mode

```
-- ASYNC example (LOG_ARCHIVE_DEST_2):
LOG_ARCHIVE_DEST_2='SERVICE=STDBY LGWR ASYNC
  VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
  DB_UNIQUE_NAME=STDBY'

-- SYNC example:
LOG_ARCHIVE_DEST_2='SERVICE=STDBY LGWR SYNC AFFIRM
  VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
  DB_UNIQUE_NAME=STDBY'
```

Things to Remember:
SYNC requires AFFIRM on the standby side — the standby must write redo to disk before acknowledging. Without AFFIRM, SYNC alone does not guarantee zero data loss because the standby might ACK before completing the disk write.

9. Protection Modes: Maximum Protection, Availability, and Performance

Oracle Data Guard offers three protection modes that define the trade-off between data protection and primary database availability. Choosing the right mode depends on your RPO (Recovery Point Objective) and business requirements.

Maximum Protection

- Zero data loss — no committed data is ever lost
- Requires SYNC, LGWR, AFFIRM on at least one standby
- If no standby can receive redo, the PRIMARY SHUTS DOWN rather than risk data loss
- Suitable for financial systems, banking cores, or any system where data loss is completely unacceptable
- Risk: if network fails, your primary goes down — plan your network carefully

Maximum Availability

- Zero data loss under normal conditions
- Requires SYNC, LGWR, AFFIRM on at least one standby
- Unlike Maximum Protection, if the standby becomes unreachable, the primary automatically downgrades to Maximum Performance mode and continues running
- This is the most balanced option — protection without risking primary availability
- Most commonly used in production environments that require zero RPO but cannot afford primary downtime due to standby issues

Maximum Performance (Default)

- Potential data loss — commits do not wait for standby
- Uses ASYNC transport
- Primary performance is not affected by standby status
- Suitable for dev/test standby or when standby is far away with high latency

Feature	Max Protection	Max Availability	Max Performance
Data Loss	Zero	Zero (normal)	Possible
Transport	SYNC	SYNC	ASYNC
Standby failure impact	Primary shuts down	Degrades gracefully	No impact
Performance impact	Yes	Yes (minor)	None

```
-- Set protection mode (run on primary, mount state):
ALTER DATABASE SET STANDBY DATABASE TO MAXIMIZE AVAILABILITY;
-- or:
ALTER DATABASE SET STANDBY DATABASE TO MAXIMIZE PERFORMANCE;
-- or:
ALTER DATABASE SET STANDBY DATABASE TO MAXIMIZE PROTECTION;
```

```
-- Verify:
SELECT PROTECTION_MODE, PROTECTION_LEVEL FROM V$DATABASE;
```

10. Standby Redo Logs Best Practices

Standby Redo Logs (SRLs) are special redo log groups on the standby that receive real-time redo from the primary. Without SRLs, real-time apply is not possible and the standby will always lag behind by at least one archived log.

SRL Sizing Rules

- SRLs must be the same size as the online redo logs on the primary
- You need one more SRL group than the number of online redo log groups on the primary
- Formula: SRL groups = (online redo log groups per thread) + 1
- For a RAC primary with 2 threads: (groups per thread + 1) x 2 threads

Creating Standby Redo Logs

```
-- Check existing online redo log sizes on primary:
SELECT GROUP#, MEMBERS, BYTES/1024/1024 AS MB FROM V$LOG;

-- Create SRLs on standby (example: 3 groups + 1 = 4 SRLs, 200MB each):
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  GROUP 11 '/u01/oradata/stdby/standby_redo_01.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  GROUP 12 '/u01/oradata/stdby/standby_redo_02.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  GROUP 13 '/u01/oradata/stdby/standby_redo_03.log' SIZE 200M;
ALTER DATABASE ADD STANDBY LOGFILE THREAD 1
  GROUP 14 '/u01/oradata/stdby/standby_redo_04.log' SIZE 200M;

-- Verify SRLs:
SELECT GROUP#, THREAD#, SEQUENCE#, ARCHIVED, STATUS, BYTES/1024/1024 MB
FROM V$STANDBY_LOG;
```

Common DBA Mistake:
If SRLs are not created or are incorrectly sized, the RFS process cannot write real-time redo. You will see gaps, delays, and the apply process will always wait for archives. This is one of the most common setup mistakes.

Production Best Practice:
Always create SRLs as part of standby database setup, before starting MRP. Drop and recreate them if you resize online redo logs on the primary. Always keep them multiplexed across different disk groups if using ASM.

11. Real-Time Apply

Real-Time Apply allows MRP to apply redo as it is received into the Standby Redo Logs, without waiting for the log to be archived first. This dramatically reduces apply lag and gets the standby as close to the primary as possible.

How Real-Time Apply Works

With SRLs in place and real-time apply enabled, the sequence is:

- LGWR on primary writes redo to online redo log
- LNS sends redo to standby RFS in real-time
- RFS writes redo to the current SRL group on standby
- MRP reads from the active SRL (USING CURRENT LOGFILE) and applies
- When primary switches logs, standby SRL is archived and MRP switches to new SRL

Enabling Real-Time Apply

```
-- Start MRP with real-time apply:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE
  USING CURRENT LOGFILE DISCONNECT FROM SESSION;

-- Confirm it's working:
SELECT PROCESS, STATUS, SEQUENCE# FROM V$MANAGED_STANDBY
WHERE PROCESS = 'MRP0';
-- STATUS should show: APPLYING_LOG

-- Check if applying from SRL or archived log:
SELECT PROCESS, STATUS, THREAD#, SEQUENCE#, BLOCK# FROM V$MANAGED_STANDBY;
-- RFS process SEQUENCE# should match current primary sequence
```

Things to Remember:
If MRP shows WAIT_FOR_LOG status even with real-time apply enabled, check if SRLs exist and are properly sized. Also verify the transport mode — ARCH transport cannot support real-time apply.

12. Log Apply Services

Log Apply Services is the umbrella term for everything on the standby side that controls how and when redo is applied. Understanding the different apply options helps in tuning apply lag and managing maintenance windows.

Apply Delay

Apply delay intentionally introduces a lag in redo apply. This is useful as a protection against human errors — if someone accidentally drops a table on the primary, you have a window before it is applied on the standby to recover from the standby.

```
-- Set apply delay of 2 hours:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DELAY 120 DISCONNECT;
-- DELAY is in minutes

-- Remove apply delay:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE NODELAY DISCONNECT;

-- Alternatively, set in LOG_ARCHIVE_DEST_n on primary:
LOG_ARCHIVE_DEST_2='SERVICE=STDBY LGWR ASYNC DELAY=120 ...'
```

Recovery to a Point in Time

During maintenance or testing, you may want to recover the standby only up to a specific SCN or time:

```
-- Recover until a specific SCN:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE UNTIL CHANGE 5000000
DISCONNECT;

-- Recover until a specific time:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE
  UNTIL TIME '2024-01-15 14:00:00' DISCONNECT;
```

Production Best Practice:
Using apply delay on one of the standby databases (if you have multiple) is a recommended safety net. It gives you a recovery window for human errors without compromising the zero-lag standby used for failover.

13. Data Guard Broker Overview

Data Guard Broker is a distributed management framework that automates the creation, maintenance, and monitoring of Data Guard configurations. It simplifies role transitions (switchover and failover) and provides a single management interface through DGMGRL (Data Guard Manager command-line interface) or Cloud Control.

Without the broker, every switchover or failover involves multiple manual steps on both primary and standby. With the broker, it becomes a single command. Broker also enables Fast-Start Failover — automatic failover without human intervention.

Broker Configuration Files

The broker configuration is stored in two binary files on the primary database host. These files are automatically replicated to the standby.

```
-- Parameter to define broker config file location:
DG_BROKER_CONFIG_FILE1 =
'/u01/app/oracle/product/19c/dbhome_1/dbs/dr1PROD.dat'
DG_BROKER_CONFIG_FILE2 =
'/u01/app/oracle/product/19c/dbhome_1/dbs/dr2PROD.dat'

-- Enable broker (set on both primary and standby):
ALTER SYSTEM SET DG_BROKER_START = TRUE;
```

Enabling and Connecting to Broker

```
-- Start DGMGRL:
dgmgrl /
-- or:
dgmgrl sys/password@PROD

-- Create broker configuration:
CREATE CONFIGURATION 'DG_PROD' AS
  PRIMARY DATABASE IS 'PROD'
  CONNECT IDENTIFIER IS PROD;

-- Add standby to configuration:
ADD DATABASE 'STDBY' AS
  CONNECT IDENTIFIER IS STDBY
  MAINTAINED AS PHYSICAL;

-- Enable the configuration:
ENABLE CONFIGURATION;

-- Check configuration status:
SHOW CONFIGURATION;
```

14. Important Initialization Parameters

Getting the initialization parameters right on both primary and standby is critical. A wrong parameter is often the root cause of transport failures, apply failures, and role transition issues.

Parameters on Primary Database

Parameter	Purpose and Recommended Setting
DB_NAME	Must be same on primary and all standbys
DB_UNIQUE_NAME	Must be unique across all databases in the DG config
LOG_MODE	Must be ARCHIVELOG on primary
LOG_ARCHIVE_DEST_1	Local archive destination
LOG_ARCHIVE_DEST_2	Standby transport destination with SERVICE, LGWR, ASYNC/SYNC
REMOTE_LOGIN_PASSWORDFILE	Must be EXCLUSIVE or SHARED
FAL_SERVER	TNS alias of primary (used by standby to request missing archives)
FAL_CLIENT	TNS alias of standby (used by primary to push archives back)
STANDBY_FILE_MANAGEMENT	Set AUTO to automatically add datafiles on standby
DB_FILE_NAME_CONVERT	Converts datafile paths from primary to standby naming
LOG_FILE_NAME_CONVERT	Converts redo log paths from primary to standby naming
ARCHIVE_LAG_TARGET	Forces log switches at intervals even if logs are not full (reduces lag)

Sample Parameter File Settings

```
-- Primary spfile settings:
*.db_name='PROD'
*.db_unique_name='PROD'
*.log_archive_dest_1='LOCATION=USE_DB_RECOVERY_FILE_DEST
  VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=PROD'
*.log_archive_dest_2='SERVICE=STDBY LGWR ASYNC
  VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
  DB_UNIQUE_NAME=STDBY'
*.log_archive_dest_state_1=ENABLE
*.log_archive_dest_state_2=ENABLE
*.remote_login_passwordfile=EXCLUSIVE
*.fal_server=STDBY
*.fal_client=PROD
*.standby_file_management=AUTO
*.archive_lag_target=1800
*.dg_broker_start=TRUE
*.dg_broker_config_file1='/u01/oracle/dbs/dr1PROD.dat'
*.dg_broker_config_file2='/u01/oracle/dbs/dr2PROD.dat'
```

```
-- Standby-specific additions:
STDBY.db_unique_name='STDBY'
STDBY.fal_server=PROD
STDBY.fal_client=STDBY
STDBY.log_archive_dest_1='LOCATION=USE_DB_RECOVERY_FILE_DEST
  VALID_FOR=(ALL_LOGFILES,ALL_ROLES) DB_UNIQUE_NAME=STDBY'
STDBY.log_archive_dest_2='SERVICE=PROD LGWR ASYNC
  VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE)
  DB_UNIQUE_NAME=PROD'
```

15. Network Configuration Best Practices

Data Guard is extremely sensitive to network quality. Transport failures, log gaps, and apply delays often trace back to network issues. Getting the network configuration right from the start saves hours of troubleshooting later.

Network Recommendations

- Dedicated network for Data Guard redo transport — do not share with application traffic if possible
- Low latency is critical for SYNC transport — measure round-trip time before choosing protection mode
- Bandwidth must accommodate peak redo generation rate — check V$SYSSTAT for redo size statistics
- Enable TCP keepalives to detect dead connections quickly
- Set SQLNET.EXPIRE_TIME in sqlnet.ora to detect stale connections (value in minutes):

```
-- In sqlnet.ora on both primary and standby:
SQLNET.EXPIRE_TIME = 2
SQLNET.RECV_TIMEOUT = 30
SQLNET.SEND_TIMEOUT = 30
```

Checking Network Bandwidth and Redo Rate

```
-- Check redo generation rate on primary:
SELECT NAME, VALUE FROM V$SYSSTAT
WHERE NAME IN ('redo size', 'redo writes', 'redo blocks written');

-- Check transport status:
SELECT DEST_ID, STATUS, TARGET, ARCHIVER, SCHEDULE,
       DESTINATION, ERROR FROM V$ARCHIVE_DEST
WHERE TARGET = 'STANDBY';
```

16. TNS and Listener Configuration

Both primary and standby need proper TNS entries to communicate. The listener on the standby must be configured to accept connections from the primary, and vice versa for role transitions.

```
-- tnsnames.ora on both primary and standby hosts:
PROD =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = primary-host)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PROD)
    )
  )

STDBY =
  (DESCRIPTION =
```

```
    (ADDRESS = (PROTOCOL = TCP)(HOST = standby-host)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = STDBY)
    )
  )
```

Things to Remember:
The STATIC_SERVICES entry in listener.ora on the standby is required when the standby is in MOUNT state. A database in MOUNT state cannot register dynamically with the listener, so you must register it statically.

```
-- listener.ora on standby (required for MOUNT state registration):
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = STDBY)
      (ORACLE_HOME = /u01/app/oracle/product/19c/dbhome_1)
      (SID_NAME = STDBY)
    )
  )
```

17. Password File Management

Data Guard uses the password file for authentication between primary and standby. If password files are out of sync, redo transport fails with authentication errors.

- The SYS password must be identical on primary and standby — both password files must match
- If you change the SYS password on the primary, copy the new password file to the standby immediately
- In Oracle 12.2+, the password file is automatically propagated to the standby when DG_BROKER_START=TRUE and broker manages the configuration
- In older versions or non-broker setups, manual copy is required after every SYS password change

```
-- Copy password file from primary to standby (Oracle 19c, without
broker):
-- On primary:
scp $ORACLE_HOME/dbs/orapwPROD oracle@standby-
host:$ORACLE_HOME/dbs/orapwSTDBY

-- Verify password file location:
SELECT VALUE FROM V$PARAMETER WHERE NAME = 'remote_login_passwordfile';

-- Check password file users:
SELECT USERNAME, SYSDBA, SYSOPER FROM V$PWFILE_USERS;
```

18. RMAN Duplication for Standby & 19. Active Duplicate Method

RMAN is the standard method to create a physical standby database. The two main approaches are duplicate from backup and active duplicate. In modern production environments, active duplicate is the preferred method because it does not require a separate backup — it uses the live primary database directly.

Prerequisites Before Running Duplicate

- Password files are in sync on both servers
- TNS entries are configured and tested (tnsping works both ways)
- Listener on standby has static registration for the standby SID
- Standby has its own spfile or init.ora with the correct parameters
- Oracle software is installed on standby host (same version and patch level as primary)
- Standby data directories exist with correct permissions

Active Database Duplicate (Recommended Method)

Active duplicate copies directly from the primary over the network. No backup needed. Oracle streams the datafiles, control file, and required archived logs directly to the standby.

```
-- Step 1: Create a minimal init.ora on standby to start in NOMOUNT:
-- File: /u01/app/oracle/product/19c/dbhome_1/dbs/initSTDBY.ora
DB_NAME=PROD
DB_UNIQUE_NAME=STDBY

-- Step 2: Start standby in NOMOUNT:
sqlplus / as sysdba
STARTUP NOMOUNT
PFILE=/u01/app/oracle/product/19c/dbhome_1/dbs/initSTDBY.ora

-- Step 3: Run RMAN active duplicate from primary:
rman TARGET sys/password@PROD AUXILIARY sys/password@STDBY

-- Step 4: Execute duplicate command:
DUPLICATE TARGET DATABASE FOR STANDBY FROM ACTIVE DATABASE
  DORECOVER
  SPFILE
    SET DB_UNIQUE_NAME='STDBY'
    SET LOG_ARCHIVE_DEST_2='SERVICE=PROD LGWR ASYNC
      VALID_FOR=(ONLINE_LOGFILES,PRIMARY_ROLE) DB_UNIQUE_NAME=PROD'
    SET FAL_SERVER='PROD'
    SET FAL_CLIENT='STDBY'
    SET STANDBY_FILE_MANAGEMENT='AUTO'
    SET LOG_FILE_NAME_CONVERT='/u01/oradata/PROD','/u01/oradata/STDBY'
    SET DB_FILE_NAME_CONVERT='/u01/oradata/PROD','/u01/oradata/STDBY'
  NOFILENAMECHECK;
```

After duplicate completes, RMAN will start MRP automatically (if DORECOVER was specified). Verify standby is applying redo:

```
-- On standby after duplication:
SELECT STATUS, INSTANCE_NAME FROM V$INSTANCE;
-- Should show: MOUNTED

SELECT PROCESS, STATUS, SEQUENCE# FROM V$MANAGED_STANDBY;
-- Should show MRP0 in APPLYING_LOG status
```

Common DBA Mistake:
Never run RMAN duplicate without verifying TNS connectivity first. A failed duplicate after 6 hours of copying data (for a large database) wastes your maintenance window. Always tnsping both service names from both hosts before starting.

20. Monitoring Physical Standby Health

Day-to-day standby monitoring comes down to watching three things: Is transport working? Is apply working? Is there any lag? Here are the most reliable views to watch.

V$DATAGUARD_STATS — The Most Useful View

```
-- Check transport lag, apply lag, and estimated failover time:
SELECT NAME, VALUE, UNIT, TIME_COMPUTED
FROM V$DATAGUARD_STATS
ORDER BY NAME;

-- Key metrics to watch:
-- transport lag  : how far behind is redo shipping
-- apply lag      : how far behind is redo apply
-- estimated startup time: how long standby open would take
```

V$MANAGED_STANDBY — Process Health

```
-- Full process status:
SELECT PROCESS, CLIENT_PROCESS, STATUS, THREAD#, SEQUENCE#,
       BLOCK#, BLOCKS, DELAY_MINS
FROM V$MANAGED_STANDBY;
```

V$ARCHIVE_DEST_STATUS — Transport Status

```
-- Check all archive destinations:
SELECT DEST_ID, STATUS, ARCHIVED_THREAD#, ARCHIVED_SEQ#,
       APPLIED_THREAD#, APPLIED_SEQ#, GAP_STATUS
FROM V$ARCHIVE_DEST_STATUS
WHERE STATUS != 'INACTIVE';
```

21. Important Monitoring Queries

Check Redo Apply Progress

```
-- Confirm how far behind standby is (run on standby):
SELECT TO_CHAR(SYSDATE,'DD-MON-YYYY HH24:MI:SS') NOW,
       MAX(SEQUENCE#) LAST_APPLIED_SEQ
FROM V$ARCHIVED_LOG
WHERE APPLIED = 'YES';

-- Compare with primary current sequence:
-- (Run on primary):
SELECT THREAD#, MAX(SEQUENCE#) CURRENT_SEQ
FROM V$ARCHIVED_LOG
GROUP BY THREAD#;
```

Check Standby Lag in Seconds

```
SELECT NAME, VALUE, DATUM_TIME FROM V$DATAGUARD_STATS
WHERE NAME IN ('transport lag','apply lag');

-- Query standby from primary (using database link):
SELECT ARCH.THREAD#, ARCH.SEQUENCE# ARCH_SEQ,
       APPL.SEQUENCE# APPL_SEQ,
       (ARCH.SEQUENCE# - APPL.SEQUENCE#) LAG
FROM
  (SELECT THREAD#, MAX(SEQUENCE#) SEQUENCE# FROM V$ARCHIVED_LOG
   WHERE DEST_ID=2 GROUP BY THREAD#) ARCH,
  (SELECT THREAD#, MAX(SEQUENCE#) SEQUENCE# FROM V$ARCHIVED_LOG
   WHERE APPLIED='YES' AND DEST_ID=2 GROUP BY THREAD#) APPL
WHERE ARCH.THREAD# = APPL.THREAD#;
```

Check for Gaps

```
-- On standby, check V$ARCHIVE_GAP:
SELECT THREAD#, LOW_SEQUENCE#, HIGH_SEQUENCE# FROM V$ARCHIVE_GAP;

-- On primary, check for missing sequences:
SELECT DEST_ID, THREAD#, LOW_SEQUENCE#, HIGH_SEQUENCE#
FROM V$ARCHIVE_GAP;
```

22. Gap Detection and Resolution

A gap occurs when the standby is missing one or more archive log sequences. This happens when network connectivity between primary and standby is lost and archives from the primary do not reach the standby. Until the gap is resolved, MRP will pause and wait.

How Gaps Get Resolved

Oracle has an automatic gap resolution mechanism through FAL (Fetch Archive Log). When MRP detects a gap, it signals the FAL process on the standby, which then requests the missing archives from FAL_SERVER (typically the primary).

```
-- Check gap on standby:
SELECT * FROM V$ARCHIVE_GAP;

-- Manually trigger FAL to fetch missing logs:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE DISCONNECT;
-- FAL will automatically request missing sequences on restart

-- Verify FAL parameters:
SHOW PARAMETER FAL;
```

Manual Gap Resolution — Copying Archives

If FAL cannot automatically resolve the gap (for example, primary archives have been deleted), you may need to manually copy the missing archives to the standby:

```
-- Step 1: Identify missing sequences on standby:
SELECT THREAD#, LOW_SEQUENCE#, HIGH_SEQUENCE# FROM V$ARCHIVE_GAP;

-- Step 2: On primary, find the archive files:
SELECT NAME FROM V$ARCHIVED_LOG
WHERE THREAD# = 1 AND SEQUENCE# BETWEEN 1000 AND 1010
AND STANDBY_DEST = 'NO';

-- Step 3: Copy archives from primary to standby arch destination:
scp /u01/arch/PROD_1_1005*.arc oracle@standby-host:/u01/arch/

-- Step 4: Register the archived logs on standby:
ALTER DATABASE REGISTER LOGFILE '/u01/arch/PROD_1_1005_12345678.arc';
```

23. Archive Log Issues

Archive log related problems are the most frequent day-to-day issues in Data Guard environments. Here are the common scenarios and how to handle them:

Archive Destination Full

```
-- Check archive destination usage:
SELECT DEST_ID, DEST_NAME, STATUS, TARGET, ARCHIVER,
       SCHEDULE, DESTINATION FROM V$ARCHIVE_DEST
WHERE STATUS != 'INACTIVE';

-- Check disk space on standby:
-- (From OS): df -h /u01/arch

-- Check DB_RECOVERY_FILE_DEST usage:
SELECT SPACE_LIMIT/1073741824 AS LIMIT_GB,
       SPACE_USED/1073741824 AS USED_GB,
       SPACE_RECLAIMABLE/1073741824 AS RECLAIMABLE_GB
FROM V$RECOVERY_FILE_DEST;
```

24. FAL_SERVER and FAL_CLIENT

FAL (Fetch Archive Log) is Oracle's automatic archive gap resolution mechanism. FAL_SERVER is the database from which missing archives are fetched, and FAL_CLIENT is the database that is requesting them (usually the standby).

```
-- On standby, verify FAL parameters:
SHOW PARAMETER FAL_SERVER;  -- Should point to primary TNS alias
SHOW PARAMETER FAL_CLIENT;  -- Should be standby's own TNS alias

-- Correct setting example:
ALTER SYSTEM SET FAL_SERVER = 'PROD' SCOPE=BOTH;
ALTER SYSTEM SET FAL_CLIENT = 'STDBY' SCOPE=BOTH;
```

Things to Remember:
FAL_SERVER must be a valid TNS alias that resolves to the primary database from the standby host. Test it: tnsping PROD from standby host. If tnsping fails, FAL will not work and gaps will never be automatically resolved.

25. Apply Lag and Transport Lag

Lag is measured in two dimensions: transport lag is how far behind the primary is in shipping redo to standby. Apply lag is how far behind the standby is in applying the received redo. Both should ideally be close to zero in a healthy setup.

```
-- Best single query for lag monitoring:
SELECT NAME, VALUE, UNIT FROM V$DATAGUARD_STATS
WHERE NAME IN ('transport lag','apply lag');

-- Detailed lag breakdown from V$ARCHIVED_LOG:
SELECT THREAD#,
       MAX(SEQUENCE#) LAST_RECEIVED,
```

```
       MAX(CASE WHEN APPLIED='YES' THEN SEQUENCE# END) LAST_APPLIED,
       MAX(SEQUENCE#) - MAX(CASE WHEN APPLIED='YES' THEN SEQUENCE# END)
LAG_SEQS
FROM V$ARCHIVED_LOG
WHERE DEST_ID = 1
GROUP BY THREAD#;
```

26. Switchover Process

Switchover is a planned role reversal — primary becomes standby and standby becomes primary. No data is lost during a switchover. It is typically done for planned maintenance (patching, hardware upgrade, DR testing).

Switchover Using DGMGRL (Broker — Recommended)

```
-- Connect to broker:
dgmgrl /
SHOW CONFIGURATION;

-- Perform switchover:
SWITCHOVER TO STDBY;

-- Verify new primary:
SHOW CONFIGURATION;
SHOW DATABASE VERBOSE STDBY;
```

Switchover Using SQL (Manual — Without Broker)

```
-- Step 1: On PRIMARY – Verify switchover is ready:
SELECT SWITCHOVER_STATUS FROM V$DATABASE;
-- Should show: TO STANDBY or SESSIONS ACTIVE

-- If SESSIONS ACTIVE, disconnect active sessions or use WITH SESSION
SHUTDOWN:
ALTER DATABASE COMMIT TO SWITCHOVER TO PHYSICAL STANDBY WITH SESSION
SHUTDOWN;

-- Step 2: Primary is now a standby – restart it:
SHUTDOWN IMMEDIATE;
STARTUP MOUNT;

-- Step 3: On OLD STANDBY – verify it's ready to be primary:
SELECT SWITCHOVER_STATUS FROM V$DATABASE;
-- Should show: TO PRIMARY or SESSIONS ACTIVE

-- Step 4: Convert standby to primary:
ALTER DATABASE COMMIT TO SWITCHOVER TO PRIMARY WITH SESSION SHUTDOWN;

-- Step 5: Open the new primary:
ALTER DATABASE OPEN;

-- Step 6: Start MRP on the new standby (old primary):
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;
```

Production Best Practice:

Always use the broker for switchover and failover in production. Manual switchover has more steps, more room for error, and no automatic validation. The broker also restores the configuration after the role change automatically.

27. Failover Process

Failover is an emergency role transition when the primary database is unavailable and cannot be recovered in a reasonable time. Unlike switchover, failover may involve some data loss (depending on protection mode and whether the standby has all redo). After failover, the old primary must be reinstated or recreated as a standby.

Failover Using DGMGRL

```
-- Connect to standby broker:
dgmgrl /

-- Immediate failover (use when primary is completely unavailable):
FAILOVER TO STDBY;

-- If broker shows warnings, force the failover:
FAILOVER TO STDBY IMMEDIATE;
```

Manual Failover (No Broker)

```
-- Step 1: On standby – Flush redo (if standby can still reach primary):
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE FINISH;

-- Step 2: If primary unreachable, activate standby directly:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
ALTER DATABASE ACTIVATE STANDBY DATABASE;

-- Step 3: Open the new primary:
ALTER DATABASE OPEN;

-- Step 4: Create a new password file, update tnsnames, etc.
-- Step 5: Old primary must be re-created as standby (RMAN duplicate from
new primary)
```

Reinstating the Old Primary After Failover

```
-- Using broker (preferred – much simpler):
dgmgrl /
REINSTATE DATABASE 'PROD';

-- The broker will automatically:
-- 1. Flashback the old primary to before the failover point
```

```
-- 2. Convert it to a standby
-- 3. Start redo apply on it
-- (Requires Flashback Database to be enabled on the old primary)
```

Things to Remember:
After a failover, the old primary CANNOT simply be mounted and joined as a standby — it has diverged from the new primary. You must either reinstate it (using flashback) or re-create it as a standby from scratch using RMAN duplicate.

28. Fast-Start Failover Concepts

Fast-Start Failover (FSFO) automates the failover process — when the primary becomes unavailable, Oracle automatically promotes the standby to primary without any human intervention. This requires the Data Guard Observer process to be running.

Key Requirements for FSFO

- Data Guard Broker must be configured and enabled
- Observer must be running (a separate process on a third host or the standby host)
- Target standby must be synchronized with the primary (lag within threshold)
- FastStartFailoverThreshold parameter defines how long to wait before triggering FSFO

```
-- Enable FSFO in DGMGRL:
dgmgrl /
ENABLE FAST_START FAILOVER;

-- Set failover threshold (seconds):
EDIT CONFIGURATION SET PROPERTY FastStartFailoverThreshold = 30;

-- Start the observer (from a separate host or session):
dgmgrl /
START OBSERVER;

-- Check FSFO status:
SHOW FAST_START FAILOVER;
```

29. Data Guard Broker Commands

Below is a practical reference of the most commonly used DGMGRL commands in production environments. Learn these — they are used daily.

```
dgmgrl /                                        -- Connect using OS auth
dgmgrl sys/password@PROD                        -- Connect to broker on PROD

-- Configuration management:
SHOW CONFIGURATION;                             -- Overall health summary
SHOW CONFIGURATION VERBOSE;                     -- Detailed status
SHOW DATABASE VERBOSE 'PROD';                   -- Primary database details
SHOW DATABASE VERBOSE 'STDBY';                  -- Standby database details

-- Edit properties:
EDIT DATABASE 'STDBY' SET PROPERTY
  LogXptMode='ASYNC';                           -- Change transport mode

EDIT DATABASE 'PROD' SET PROPERTY
  RedoRoutes='(LOCAL:STDBY ASYNC)';             -- Set redo routes

-- Status commands:
SHOW ALL;                                       -- All configurable properties
SHOW INSTANCE 'PROD' ON DATABASE 'PROD';        -- RAC instance details

-- Role transitions:
SWITCHOVER TO 'STDBY';
FAILOVER TO 'STDBY';
FAILOVER TO 'STDBY' IMMEDIATE;
REINSTATE DATABASE 'PROD';

-- Enable/disable:
ENABLE CONFIGURATION;
DISABLE CONFIGURATION;
ENABLE DATABASE 'STDBY';
DISABLE DATABASE 'STDBY';

-- Observer:
START OBSERVER;
STOP OBSERVER;
SHOW OBSERVER;

-- Remove from configuration:
REMOVE DATABASE 'STDBY' PRESERVE DESTINATIONS;
REMOVE CONFIGURATION;
```

30. Common Production Issues

Here are the most frequently encountered problems in Data Guard environments, along with a structured approach to diagnosing and resolving them.

Issue 1: MRP Stopped — Not Applying Redo

```
-- Symptom: V$MANAGED_STANDBY shows no MRP0 process
-- Check alert log for errors:
-- $ORACLE_BASE/diag/rdbms/stdby/STDBY/trace/alert_STDBY.log

-- Check if MRP is running:
SELECT PROCESS, STATUS FROM V$MANAGED_STANDBY WHERE PROCESS='MRP0';

-- Restart MRP:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;
```

Issue 2: Transport Error — Cannot Connect to Standby

```
-- Symptom: V$ARCHIVE_DEST shows STATUS=ERROR for standby destination
-- Check error message:
SELECT DEST_ID, STATUS, ERROR FROM V$ARCHIVE_DEST WHERE DEST_ID=2;

-- Test TNS from primary:
tnsping STDBY

-- Test connection from RMAN or sqlplus:
sqlplus sys/password@STDBY as sysdba

-- Re-enable the destination after fixing:
ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=ENABLE;
```

Issue 3: High Apply Lag

```
-- Check lag:
SELECT NAME, VALUE FROM V$DATAGUARD_STATS WHERE NAME='apply lag';

-- Common causes:
-- 1. I/O bottleneck on standby
-- 2. Standby datafiles on slow storage
-- 3. DB_BLOCK_CHECKSUM enabled (disable on standby for performance)
-- 4. Missing SRLs causing archival wait

-- Check MRP parallel apply (19c+):
SHOW PARAMETER RECOVERY_PARALLELISM;
ALTER SYSTEM SET RECOVERY_PARALLELISM = 8;  -- Increase parallel apply
workers
```

31. ORA Errors Commonly Seen in Data Guard

ORA-16724: The standby database is not receiving redo data from the primary database.

ORA-16014: Log transport failed because the archiving destination is full or unavailable.

ORA-16191: Primary database rejected log transport because the standby password file is mismatched.

ORA-16596: The database instance is not part of the Data Guard Broker configuration.

ORA-16810: Multiple errors or warnings are detected for the specified database member.

ORA-16816: The actual database role does not match its Data Guard Broker configuration.

ORA-16826: The apply lag has exceeded the maximum threshold specified by the administrator.

ORA-16766: Redo Apply is unexpectedly stopped on the physical standby database.

ORA-01093: A switchover failed because active user sessions prevented the database from closing.

ORA-01102: Cannot mount database because another instance is already using the name.

32. Troubleshooting Real-Time Examples

Scenario 1: Standby Fell Behind After Network Outage

The network between primary and standby was down for 2 hours. Now it's back up but the standby is 2 hours behind and showing gaps.

```
-- Step 1: Verify gaps on standby:
SELECT * FROM V$ARCHIVE_GAP;
```

```
-- Step 2: Check FAL is configured:
SHOW PARAMETER FAL;

-- Step 3: Restart MRP to trigger automatic FAL resolution:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;

-- Step 4: Watch apply progress:
SELECT PROCESS, STATUS, SEQUENCE#, THREAD# FROM V$MANAGED_STANDBY;

-- Step 5: Monitor lag reducing:
SELECT NAME, VALUE FROM V$DATAGUARD_STATS WHERE NAME='apply lag';
```

Scenario 2: Standby Archive Destination Quota Exceeded

Alert log shows ORA-16401 and redo transport has stopped. Disk is full on standby archive area.

```
-- Step 1: Check disk usage from OS:
df -h /u01/arch

-- Step 2: Check RMAN archivelog backup and delete applied logs:
rman target /
DELETE ARCHIVELOG ALL COMPLETED BEFORE 'SYSDATE-1' BACKED UP 1 TIMES TO
DISK;

-- Step 3: Verify applied logs can be deleted:
SELECT MAX(SEQUENCE#) FROM V$ARCHIVED_LOG WHERE APPLIED='YES';

-- Step 4: After freeing space, re-enable transport:
ALTER SYSTEM SET LOG_ARCHIVE_DEST_STATE_2=ENABLE;

-- Step 5: Monitor transport recovery:
SELECT DEST_ID, STATUS, ERROR FROM V$ARCHIVE_DEST WHERE DEST_ID=2;
```

Scenario 3: Broker Showing ORA-16820 (Observer Not Running)

```
-- Symptom: SHOW CONFIGURATION shows Warning: ORA-16820
dgmgrl /
SHOW FAST_START FAILOVER;

-- Start observer on observer host:
dgmgrl sys/password@PROD
START OBSERVER FILE=/u01/app/oracle/observer.dat;

-- Verify observer is connected:
SHOW OBSERVER;
```

33. Recovery Scenarios

Recovering from a Corrupted Datafile on Standby

If a standby datafile becomes corrupted or accidentally deleted, it must be restored before MRP can continue. Since MRP cannot apply redo to a corrupted file, it will error out and stop.

```
-- Step 1: Cancel MRP:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

-- Step 2: Restore the corrupted file using RMAN:
rman target /
RESTORE DATAFILE 15;
-- or if using a specific backup:
RESTORE DATAFILE '/u01/oradata/STDBY/users01.dbf';

-- Step 3: Resume MRP:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;
```

Recovering a Dropped Table Using Standby

```
-- Scenario: Table was accidentally dropped on primary at 14:30
-- Standby has apply delay of 2 hours — standby is at 12:30

-- Step 1: Cancel MRP on the standby with apply delay:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;

-- Step 2: Open standby read-only (to export the table):
ALTER DATABASE OPEN READ ONLY;

-- Step 3: Export the table using expdp:
expdp system/password@STDBY TABLES=SCHEMA.DROPPED_TABLE
  DIRECTORY=DATA_PUMP_DIR DUMPFILE=recover_table.dmp

-- Step 4: Import to primary:
impdp system/password@PROD TABLES=SCHEMA.DROPPED_TABLE
  DIRECTORY=DATA_PUMP_DIR DUMPFILE=recover_table.dmp
TABLE_EXISTS_ACTION=REPLACE

-- Step 5: Return standby to recovery mode:
ALTER DATABASE CLOSE;
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;
```

34. Standby Database Maintenance

Adding a Datafile on Primary — Impact on Standby

When a new datafile is added to the primary, STANDBY_FILE_MANAGEMENT=AUTO ensures it is automatically created on the standby when that redo is applied. If set to MANUAL, MRP will pause and require manual intervention.

```
-- Verify STANDBY_FILE_MANAGEMENT on standby:
SHOW PARAMETER STANDBY_FILE_MANAGEMENT;
-- Should be AUTO

-- If MRP paused due to missing datafile:
-- On standby, create the placeholder file:
ALTER DATABASE CREATE DATAFILE 16 AS
  '/u01/oradata/STDBY/users02.dbf';

-- Then restart MRP:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;
```

35. Patch and Upgrade Considerations

Applying patches and upgrades in a Data Guard environment requires a rolling approach. The goal is to minimize or eliminate downtime by patching the standby first, then doing a quick switchover to bring the newly patched standby into production.

Rolling Patch with Physical Standby

- Step 1: Stop MRP on standby
- Step 2: Patch the standby Oracle home (apply quarterly RU/RUR)
- Step 3: Start standby with the new Oracle home — restart MRP to catch up
- Step 4: Perform a switchover — standby becomes primary (no downtime for application, very brief for connection)
- Step 5: Patch the old primary (now standby)
- Step 6: Optionally switchover back or leave the new arrangement

```
-- Before patching standby, verify MRP is active:
SELECT PROCESS, STATUS FROM V$MANAGED_STANDBY WHERE PROCESS='MRP0';

-- Cancel MRP before shutting down for patching:
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE CANCEL;
SHUTDOWN IMMEDIATE;

-- Apply patch using OPatch on the standby Oracle home:
opatch apply /patches/p35319490_190000_Linux-x86-64

-- Start standby with patched home, resume MRP:
startup mount
ALTER DATABASE RECOVER MANAGED STANDBY DATABASE USING CURRENT LOGFILE
DISCONNECT;

-- Wait for standby to be fully current, then switchover:
SELECT NAME, VALUE FROM V$DATAGUARD_STATS WHERE NAME='apply lag';
```

36. Backup Strategy in Data Guard Environment

One of the operational advantages of Data Guard is the ability to offload backups from the primary to the standby. This reduces I/O and CPU load on the production primary while still producing valid, usable backups.

RMAN Backup from Standby

```
-- Connect RMAN to the standby with catalog (or nocatalog):
rman TARGET sys/password@STDBY CATALOG rman_user/password@CATDB

-- Take a full database backup from standby:
BACKUP DATABASE PLUS ARCHIVELOG DELETE INPUT;

-- Important: Tell RMAN to use primary DB_UNIQUE_NAME for backup location:
CONFIGURE DB_UNIQUE_NAME 'PROD' CONNECT IDENTIFIER 'PROD';

-- Backup can be used to recover PRIMARY even though taken on STANDBY
-- (Both must share the same DBID — they do, since standby = copy of
primary)
```

Production Best Practice:
Always register all backups with an RMAN catalog when backing up from a standby. The catalog tracks which database a backup belongs to using DBID. Without a catalog, recovery from standby backups can be complicated by path and DBID confusion.

37. RMAN Best Practices

- Configure RMAN backup retention policy consistently on both primary and standby
- Use RMAN catalog — especially mandatory in DG environments with multi-standby setups
- Test recovery from backup regularly — take a backup, restore to a test system quarterly
- Backup controlfile separately: BACKUP CURRENT CONTROLFILE;
- Include ARCHIVELOG in every backup strategy to cover redo for point-in-time recovery

```
-- Recommended RMAN configuration for Data Guard environments:
CONFIGURE DEFAULT DEVICE TYPE TO DISK;
CONFIGURE BACKUP OPTIMIZATION ON;
CONFIGURE CONTROLFILE AUTOBACKUP ON;
CONFIGURE RETENTION POLICY TO RECOVERY WINDOW OF 7 DAYS;
CONFIGURE ARCHIVELOG DELETION POLICY TO APPLIED ON STANDBY;
-- The last line tells RMAN not to delete archives until they are applied
on standby
```

38. Performance Tuning in Data Guard

Common Performance Issues and Solutions

Issue	Solution
High apply lag	Increase RECOVERY_PARALLELISM. Move standby datafiles to faster storage.
LGWR slow due to SYNC	Move standby closer (reduce network latency). Switch to ASYNC if RPO allows.
RFS process bottleneck	Tune network buffer sizes. Add SRL groups.
MRP waiting for I/O	Use ASM with faster disk groups. Stripe standby redo logs across disks.
Excessive archiving on standby	Increase DB_RECOVERY_FILE_DEST_SIZE or archive to faster disk.

```
-- Check MRP parallel processes (19c+):
SELECT COUNT(*) FROM V$MANAGED_STANDBY WHERE PROCESS LIKE 'PR%';

-- Increase parallel apply workers:
ALTER SYSTEM SET RECOVERY_PARALLELISM = 8 SCOPE=BOTH;

-- Check I/O wait on standby:
SELECT EVENT, TOTAL_WAITS, TIME_WAITED FROM V$SYSTEM_EVENT
WHERE EVENT LIKE '%IO%' ORDER BY TIME_WAITED DESC;
```

39. Common Mistakes DBAs Make

Common DBA Mistake:
Not configuring Standby Redo Logs. This is the single most common setup mistake. Without SRLs, real-time apply is impossible and the standby always lags behind by at least one log file.

Common DBA Mistake:
Using ARCH transport for a production standby that needs near-zero data loss. ARCH transport introduces lag equal to the time between log switches. If the primary crashes mid-log, that entire log is lost.

Common DBA Mistake:
Not testing failover before disaster strikes. A failover that has never been tested will almost certainly have surprises — DNS entries, application connection strings, VIP reassignment. Test quarterly.

Common DBA Mistake:
Setting STANDBY_FILE_MANAGEMENT=MANUAL on standby. When the primary adds a new datafile or tablespace, MRP stops and waits indefinitely on the standby. Always use AUTO.

Common DBA Mistake:
Forgetting to copy the password file after changing SYS password. Redo transport uses SYS authentication. If password files don't match, transport fails immediately with an authentication error.

Common DBA Mistake:
Not registering RMAN catalog when backing up from standby. Without a catalog, RMAN cannot correlate the standby backup with primary recovery. You may be unable to use the backup for primary recovery.

Common DBA Mistake:
Deleting archived logs on primary before they are applied on standby. Use CONFIGURE ARCHIVELOG DELETION POLICY TO APPLIED ON STANDBY in RMAN to prevent this.

40. Important Production Checklists

Standby Setup Checklist

- DB_NAME is same on primary and standby
- DB_UNIQUE_NAME is unique for each database
- REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE
- Password files are synchronized
- tnsping works both ways between primary and standby
- Static listener registration on standby (for MOUNT state)
- STANDBY_FILE_MANAGEMENT=AUTO on standby
- FAL_SERVER and FAL_CLIENT are correctly set on both databases
- Standby Redo Logs created (n+1 groups, same size as online redo logs)
- LOG_ARCHIVE_DEST_n correctly configured with LGWR, ASYNC/SYNC, DB_UNIQUE_NAME
- DG_BROKER_START=TRUE on both databases
- Broker configuration created and enabled
- MRP running with USING CURRENT LOGFILE (real-time apply)
- RMAN ARCHIVELOG DELETION POLICY set to APPLIED ON STANDBY

41. Daily Health Check Activities

A healthy Data Guard environment requires daily monitoring. Here is a checklist of what to verify every day:

```
-- Daily DG Health Check Script:
-- Run on primary:

-- 1. Check protection mode and database role:
SELECT DB_UNIQUE_NAME, OPEN_MODE, PROTECTION_MODE,
       DATABASE_ROLE FROM V$DATABASE;

-- 2. Check transport destinations:
SELECT DEST_ID, STATUS, TARGET, ARCHIVER, SCHEDULE, ERROR
FROM V$ARCHIVE_DEST_STATUS
WHERE STATUS != 'INACTIVE';

-- 3. Check current sequence on primary:
SELECT THREAD#, MAX(SEQUENCE#) FROM V$ARCHIVED_LOG
WHERE DEST_ID=1 GROUP BY THREAD#;

-- Run on standby:

-- 4. Check MRP status:
SELECT PROCESS, STATUS, SEQUENCE#, THREAD# FROM V$MANAGED_STANDBY;

-- 5. Check lag:
SELECT NAME, VALUE FROM V$DATAGUARD_STATS
```

```
WHERE NAME IN ('transport lag','apply lag');

-- 6. Check for gaps:
SELECT * FROM V$ARCHIVE_GAP;

-- 7. Check broker status (from DGMGRL):
-- SHOW CONFIGURATION;

-- 8. Check alert logs on both primary and standby for ORA- errors:
-- tail -100
$ORACLE_BASE/diag/rdbms/<db_unique_name>/<sid>/trace/alert_<sid>.log
```

42. DR Drill Planning

A DR drill validates that your Data Guard setup actually works in a failover scenario. Many organizations discover configuration errors only during an actual disaster — which is the worst possible time. DR drills prevent that.

DR Drill Using Switchover (Preferred — No Data Loss)

- Step 1: Notify application and middleware teams — the database will change roles temporarily
- Step 2: Verify standby is current (zero lag) before starting
- Step 3: Perform switchover using DGMGRL: SWITCHOVER TO STDBY
- Step 4: Verify new primary is open and accessible: application teams confirm connectivity
- Step 5: Run basic application tests on new primary
- Step 6: Switchover back to original primary
- Step 7: Document results — include timings, issues encountered, and resolution steps

DR Drill Checklist

- Alert log on both databases checked before drill for pre-existing errors
- MRP confirmed active with zero lag before switchover
- Application connection strings tested after switchover
- DNS / VIP / SCAN listener updated if required
- Monitoring tools updated to point to new primary
- Broker configuration verified after drill completes
- Lag returns to zero after switchback

Production Best Practice:
DR drills should be done at least quarterly. Use switchover so there is no data loss and minimal disruption. Document every step and every problem you find. Each drill makes your team faster and more confident for an actual event.

43. Best Practices in Real Production Environments

- Use Data Guard Broker — always. Manual Data Guard management is error-prone and harder to maintain
- Run at Maximum Availability mode for mission-critical databases
- Keep standby database version exactly in sync with primary — patch both together
- Use ASM for both primary and standby datafile management — it simplifies storage management
- Monitor V$DATAGUARD_STATS daily — automate alerts when lag exceeds thresholds
- Enable Flashback Database on both primary and standby — enables reinstate after failover
- Use RMAN catalog — mandatory in DG environments for proper backup tracking
- Test FAL resolution periodically — manually create a small gap and watch it auto-resolve
- Do not run heavy batch jobs on the standby host when using Active Data Guard — it competes with MRP

- Set ARCHIVE_LAG_TARGET to force frequent log switches and keep standby closer to primary

```
-- Enable Flashback Database (recommended on both primary and standby):
ALTER SYSTEM SET DB_FLASHBACK_RETENTION_TARGET=1440;  -- 24 hours
ALTER DATABASE FLASHBACK ON;

-- Set archive lag target (60 minutes = 3600 seconds, set in seconds):
-- Note: ARCHIVE_LAG_TARGET is in seconds
ALTER SYSTEM SET ARCHIVE_LAG_TARGET = 1800 SCOPE=BOTH;  -- 30 minutes
```

44. Interview-Focused Practical Concepts

These are the questions most commonly asked in Oracle DBA interviews relating to Data Guard. The answers here are practical, not textbook.

Q: What is the difference between switchover and failover?

Switchover is planned — both sides participate, no data is lost, the old primary becomes a standby. Failover is unplanned — the primary is unavailable, the standby is activated, possible data loss depending on protection mode, and the old primary must be reinstated or recreated.

Q: What happens if you use SYNC transport without AFFIRM?

SYNC means the primary waits for the standby to ACK before committing. But without AFFIRM, the standby ACKs as soon as it receives the redo in memory — not when it writes to disk. This means if the standby crashes between ACK and disk write, the data is gone. True zero data loss requires SYNC + AFFIRM.

Q: How many SRL groups should you create?

One more than the number of online redo log groups per thread on the primary. For example, if the primary has 3 online redo log groups per thread, create 4 SRL groups per thread on the standby.

Q: What is FAL and when does it trigger?

FAL (Fetch Archive Log) is Oracle's automatic gap resolution mechanism. It triggers when MRP detects a gap in archive log sequences. The standby's FAL process sends a request to FAL_SERVER (usually the primary) to ship the missing archives. It is completely automatic as long as FAL_SERVER and FAL_CLIENT are correctly set.

Q: What is the role of V$DATAGUARD_STATS?

This view provides real-time lag metrics — transport lag, apply lag, and estimated failover time. It is the most useful single view for quickly assessing the health of a Data Guard configuration.

Q: Can you take backups from the standby?

Yes. RMAN can connect to the standby and take a full database backup. Because the standby has the same DBID as the primary, these backups can be used to recover the primary. Using a catalog is recommended to properly track these backups.

45. Final Summary and Key Takeaways

Oracle Data Guard physical standby is one of the most powerful and reliable high availability solutions in the enterprise database world. When configured correctly and monitored consistently, it is a near-invisible safety net that protects your data 24 hours a day.

The 10 Things You Must Never Forget

- Always configure Standby Redo Logs — real-time apply is impossible without them
- Keep password files in sync whenever SYS password changes
- Use LGWR transport in production — ARCH transport introduces unnecessary lag
- Broker is not optional in real production environments — use it
- STANDBY_FILE_MANAGEMENT must be AUTO
- Enable Flashback Database — it makes reinstate after failover possible without re-duplication
- Set CONFIGURE ARCHIVELOG DELETION POLICY TO APPLIED ON STANDBY in RMAN
- Monitor lag every day — automate it with a script or OEM alert
- Test failover quarterly — DR that has never been tested is not DR
- Keep the standby on the same patch level as the primary

Quick Reference: Most Used Queries in Production

```
-- 1. Overall DG health (run on standby):
SELECT NAME, VALUE FROM V$DATAGUARD_STATS;

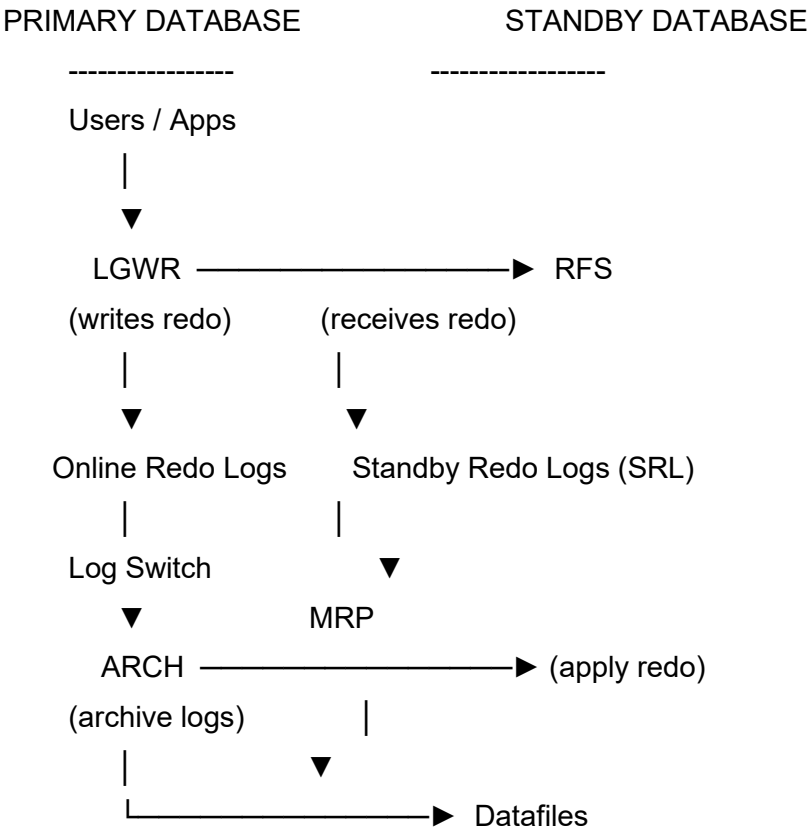
-- 2. MRP status:
SELECT PROCESS, STATUS, SEQUENCE# FROM V$MANAGED_STANDBY;

-- 3. Gap check:
SELECT * FROM V$ARCHIVE_GAP;

-- 4. Transport destination status (run on primary):
SELECT DEST_ID, STATUS, ERROR FROM V$ARCHIVE_DEST WHERE TARGET='STANDBY';

-- 5. Database role and mode:
SELECT DB_UNIQUE_NAME, DATABASE_ROLE, OPEN_MODE,
       PROTECTION_MODE FROM V$DATABASE;

-- 6. Broker quick status:
-- dgmgrl / <<< 'SHOW CONFIGURATION'
```



👉 Flow:

- LGWR → real-time redo
- ARCH → after log switch
- RFS writes → SRL / Archive
- MRP applies redo

✅ 🔷 **Data Guard Protection Modes**

1. Maximum Protection

- Zero data loss
- Uses **SYNC + AFFIRM**
- Primary shuts down if standby unavailable

2. Maximum Availability ✅ (most used)

- Zero data loss (normal condition)
- Uses **SYNC**
- Falls back to performance if standby fails

3. Maximum Performance ✅ (default)

- Possible data loss
- Uses **ASYNC**
- Best performance

✅ 🔷 **Types of Standby Databases**

1. Physical Standby

- Block-level copy
- Uses Redo Apply
- ✅ Most used

2. Logical Standby

- Applies SQL statements
- Can be open for read/write

3. Snapshot Standby

- Temporary RW standby
- Used for testing
- Can revert back

Oracle Data Guard: Key Configuration Points Every DBA Must Know

- Create **Standby Redo Logs (SRL)**
- SRL = **Online Redo Logs + 1 (per thread)**
- Use **LGWR transport (not ARCH-only)**
- Use **SYNC + AFFIRM for zero data loss**
- Monitor **Transport Lag / Apply Lag**
- Set **STANDBY_FILE_MANAGEMENT = AUTO**
- Configure **FAL_SERVER / FAL_CLIENT**
- Ensure **TNS connectivity both sides**
- Keep **password file same**
- Ensure **MRP running (APPLYING_LOG)**

Core Flow

LGWR → LNS → RFS → SRL → MRP → Datafiles